

Report II

Solving the system of linear equations using LU decomposition

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1. Introduction

In many numerical applications it is necessary to construct a function that satisfies a set of predefined conditions at certain points. These points are called collocation nodes. We define basis of functions such that each of them must fulfill set of conditions cast on the function values or calculated derivatives.

A convenient choice for such basis functions is a polynomial, because polynomials are easy to differentiate, integrate and evaluate numerically.

A polynomial of degree N can be written as a linear combination of monomials:

$$f(x) = \sum_{i=0}^N a_i x^i$$

where a_i are unknown coefficients.

In this exercise we construct a 5th-degree polynomial

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5$$

whose coefficients are determined by imposing $(N + 1)$ six conditions:

- function value at four nodes
- derivative value at two nodes

These conditions form a system of six linear equations, which can be solved numerically using LU decomposition from the LAPACK library.

2. Description of the Numerical Task

The polynomial coefficients are determined from the following conditions:

- $f(x_0) = 0$
- $f'(x_0) = 1$
- $f(x_1) = 1$
- $f(x_2) = 0$
- $f(x_3) = 1$
- $f'(x_3) = 0$

The polynomial derivative can be calculated as

$$f'(x) = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4$$

Substituting the node values into the polynomial and its derivative produces a system of linear equations

$$A\mathbf{a} = \mathbf{b}$$

where:

- A - matrix built from powers of x
- $\mathbf{a}$ - vector of unknown coefficients
- $\mathbf{b}$ - vector of imposed conditions

The nodes used in the program are

$$x_0 = 0, x_1 = 1, x_2 = 2, x_3 = 3$$

and the condition vector is

$$\mathbf{b} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}$$

3. Numerical method

The system of equations is solved using the LU decomposition.

LU decomposition factorizes matrix A into

$$A = LU$$

where:

- L - lower triangular matrix
- U - upper triangular matrix

This decomposition allows the linear system to be solved efficiently.

The implementation uses LAPACK routines:

- `dgetrf()` - LU factorization of the matrix
- `dgetrs()` - solving the system using the LU decomposition

Before solving, the determinant of the matrix is checked to ensure the system has a unique solution. If it is different than zero we shall look for a solution.

4. Implementation

The program is written in C++ and uses the LAPACK library.

The main steps of the program are:

- 1. Construction of matrix A**

The matrix is built from powers of the node coordinates and derivatives.

- 2. Defining the right-hand side vector b**

$b = \{0.0, 1.0, 1.0, 0.0, -1.0, 0.0\}$

- 3. Performing the LU decomposition**

`LAPACKE_dgetrf`

- 4. Solving the linear system**

`LAPACKE_dgetrs`

5. Storing the polynomial coefficients

The coefficients are written to the text file *coefficients.txt*

6. Generating points for plotting

Values of the polynomial are calculated for

$$x \in [0, 3]$$

with a step

$$dx = 0.01$$

and stored in *plot.txt*

These values can be used to generate the graph of a polynomial.

5. Results

The computed coefficients define the polynomial:

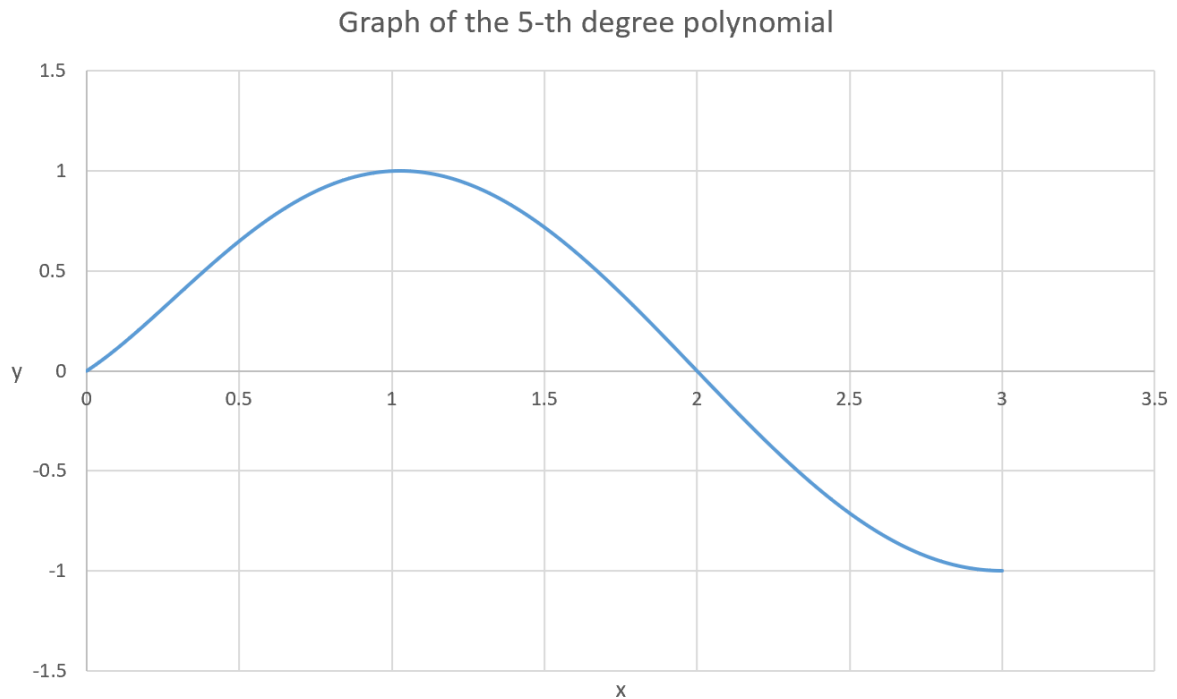
$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5$$

From the numerical solution obtained by the program the polynomial is equal to

$$f(x) = x + 1.5x^2 - 2.148148x^3 + 0.722222x^4 - 0.074074x^5$$

This polynomial satisfies all imposed constraints at the nodes.

The generated data file *plot.txt* can be used to visualize the polynomial and confirm that the curve passes through the required points with the specified slopes.



5. Conclusions

In this exercise a 5th-degree polynomial satisfying six conditions was constructed by solving a system of linear equations.

The coefficients were obtained using LU decomposition from the LAPACK library, which is a reliable and efficient method for solving linear systems.

The program successfully:

- constructed the system matrix
- solved the linear system
- generated polynomial coefficients
- produced data for visualization using a graph